

Towards fast and constant-time implementations of Hamming Quasi-Cyclic

A case study on ARM Cortex-M4 devices

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Overview

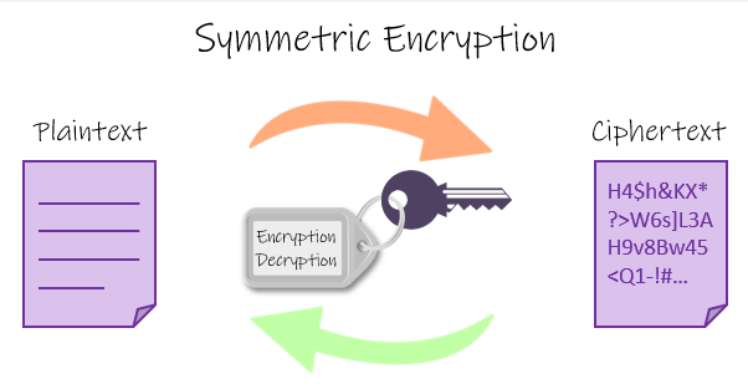
1. Introduction
2. Background
   - Polynomial Multplication
   - Coding theory
   - Hamming Quasi-Cyclic
3. Implementation
   - Polynomial multiplication
4. Experimental results
5. Conclusions

Introduction

Kerchoff's Principle

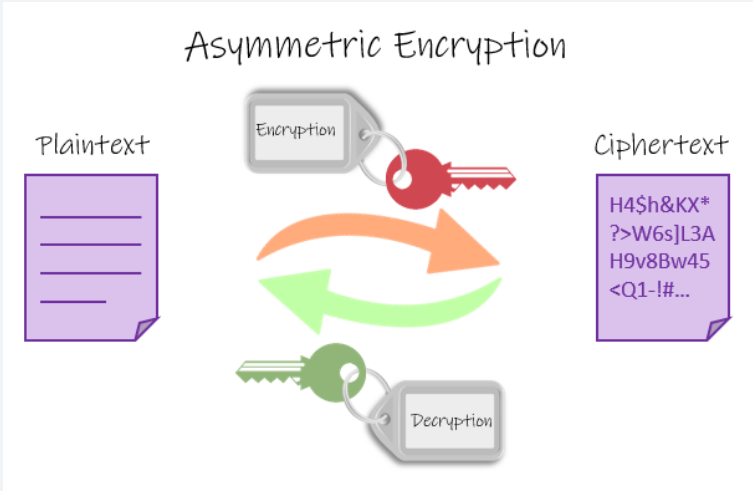
Everything about a cryptosystem is publicly known, **except for the key**

Symmetric encryption



Key: a n-bit string

Asymmetric encryption



Key: the product of two prime numbers, *i.e.* $n = p \cdot q$

Cryptanalytic attacks

What are the best attacks on modern cryptosystems?

On Classical Computers

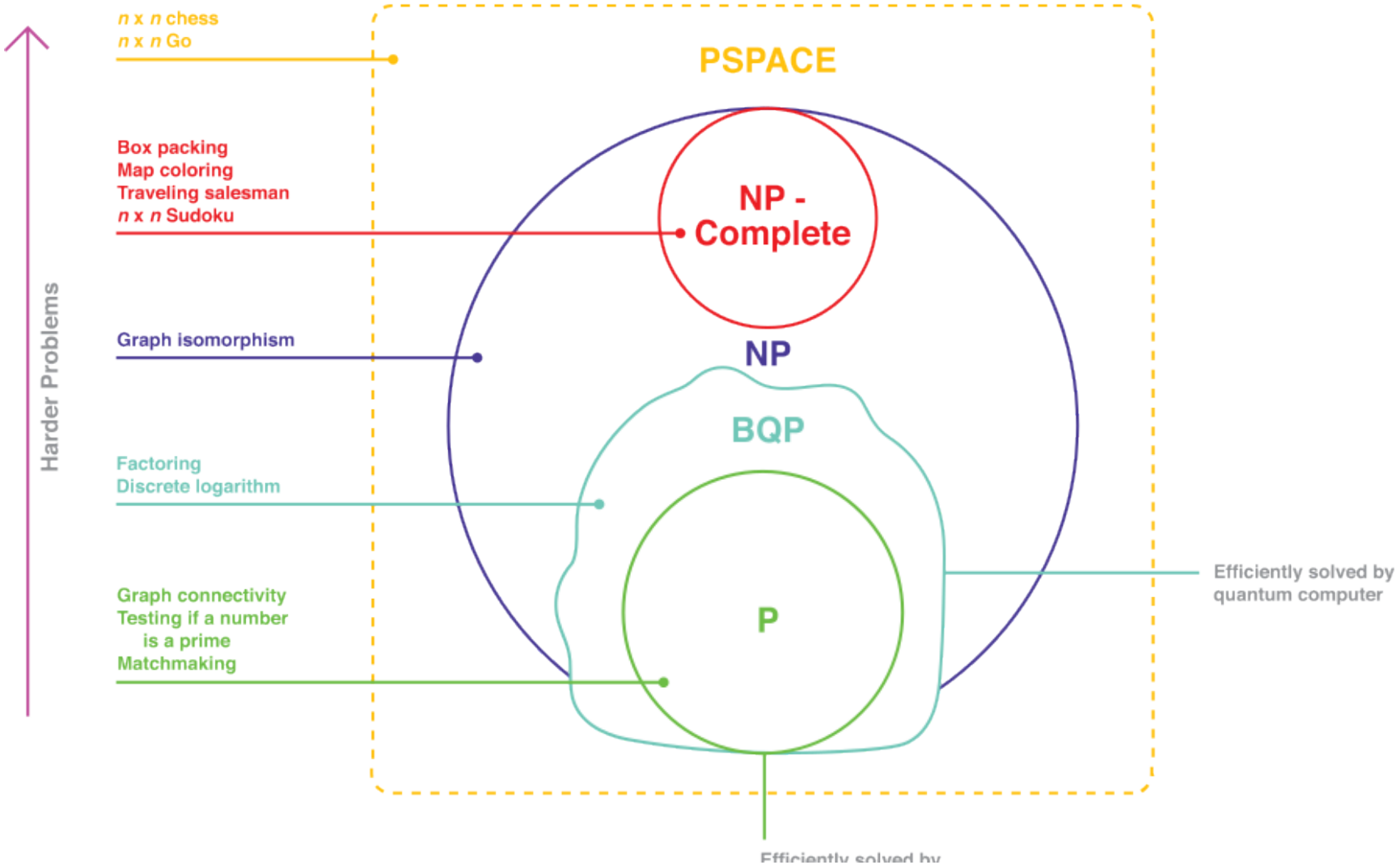
- Symmetric: exhaustive search, $\mathcal{O}(2^n)$
- Asymmetric: General Number Field Sieve, $\approx \mathcal{O}(2^{n-\epsilon})$

On Quantum Computers

- Symmetric: Grover's algorithm, $\mathcal{O}(\sqrt{2^n})$
- Asymmetric: Shor's algorithm, $\approx \mathcal{O}((\log n)^2)$

Shor's algorithm can break asymmetric cryptosystems in polynomial time!

The Computational Complexity Zoo



NIST Post-Quantum Cryptography Standardization

Timeline

- 26/02/16: Announcement of NIST's call for submissions
- 21/12/17: Announcement of the first round candidates (69)
- 30/01/19: Announcement of the second round candidates (26)
- 22/07/20: Announcement of the third round candidates (7 finalists, 8 alternatives)
- 05/07/22: Announcement of the fourth round candidates (4)
- 2022-2024: Publication of the standard drafts

Standardization candidates

Third Round winner

- CRYSTALS-Kyber: lattice-based

NIST decided to standardize at least another PQC scheme, in case lattice-based crypto fails

Fourth Round candidates

- ~~**S**upersingular **I**sogeny **K**ey **E**ncapsulation~~
- **B**it Fl**i**pping **K**ey **E**ncapsulation [ABB+22]
- **H**amming **Q**uasi-**C**yclic [MAB+18]
- Classic McEliece [BCC+22]

BIKE, Classic McEliece and HQC are all code-based cryptosystems.

Overview

The target of this thesis is Hamming Quasi-Cyclic

Goals

- High performance implementation for Cortex-M4
- Protection against timing attacks [GHJ+22, WBB+22]
- Protection against power attacks [SRSW20, SHR+22]

Roadmap

- Evaluate various polynomial multiplication algorithms
- Assess the constant-time behavior of all components
- Implement a masking scheme for the PKE primitives
- Compare the performance with the other Round 4 candidates

Background

Polynomial multiplication 101

$$a(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$$
$$b(x) = b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n$$

Our goal: calculate:

$$c(x) = a(x) \cdot b(x) = c_0 + c_1 x + c_2 x^2 + \cdots + c_{2n} x^{2n}$$

- Schoolbook multiplication
- Karatsuba algorithm
- FFT-based approach

Schoolbook multiplication

Multiply each coefficient of $a(x)$ with each coefficient of $b(x)$

$$\begin{aligned} c(x) =& a(x) \cdot b(x) \\ =& a_0 \cdot (b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n) + \\ & + a_1 x \cdot (b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n) + power \\ & + \cdots + \\ & + a_n x^n \cdot (b_0 + b_1 x + b_2 x^2 + \cdots + b_n x^n) \end{aligned}$$

- n^2 multiplications
- n^2 additions

Overall complexity: $\mathcal{O}(n^2)$

Karatsuba multiplication

Recursively split $a(x)$ and $b(x)$ into two halves

$$a(x) = a_0(x) + x^m a_1(x) \qquad b(x) = b_0(x) + x^m b_1(x)$$

$$\begin{aligned} c(x) &= a(x)b(x) \\ &= (a_0(x) + x^m a_1(x))(b_0(x) + x^m b_1(x)) \\ &= a_1(x)b_1(x)x^{2m} + (a_1(x)b_0(x) + a_0(x)b_1(x))x^m + a_0(x)b_0(x) \\ &= z_2(x)x^{2m} + z_1(x)x^m + z_0(x) \end{aligned}$$

Trick: rewrite $z_1(x)$ as $(a_0(x) + a_1(x))(b_0(x) + b_1(x)) - z_0(x) - z_2(x)$

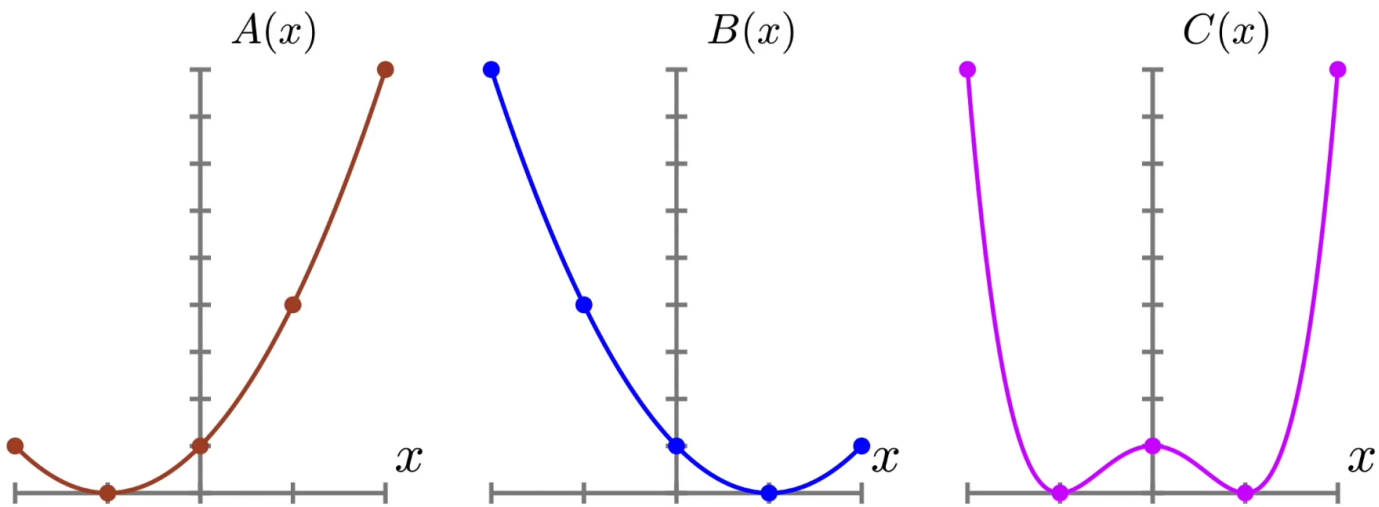
Overall complexity: $\mathcal{O}(n^{\log_2 3})$ if $m \approx n/2$

Sample and interpolate

- Sample $a(x)$ and $b(x)$ at $2n+1$ points
- Multiply each pair of samples
- Interpolate the points to get $c(x)$

$$A(x) = x^2 + 2x + 1 \qquad B(x) = x^2 - 2x + 1 \qquad C(x) = x^4 - 2x^2 + 1$$

$$[(-2,1),(-1,0),(0,1),(1,4),(2,9)] \qquad [(-2,9),(-1,4),(0,1),(1,0),(2,1)] \qquad [(-2,9),(-1,0),(0,1),(1,0),(2,9)]$$



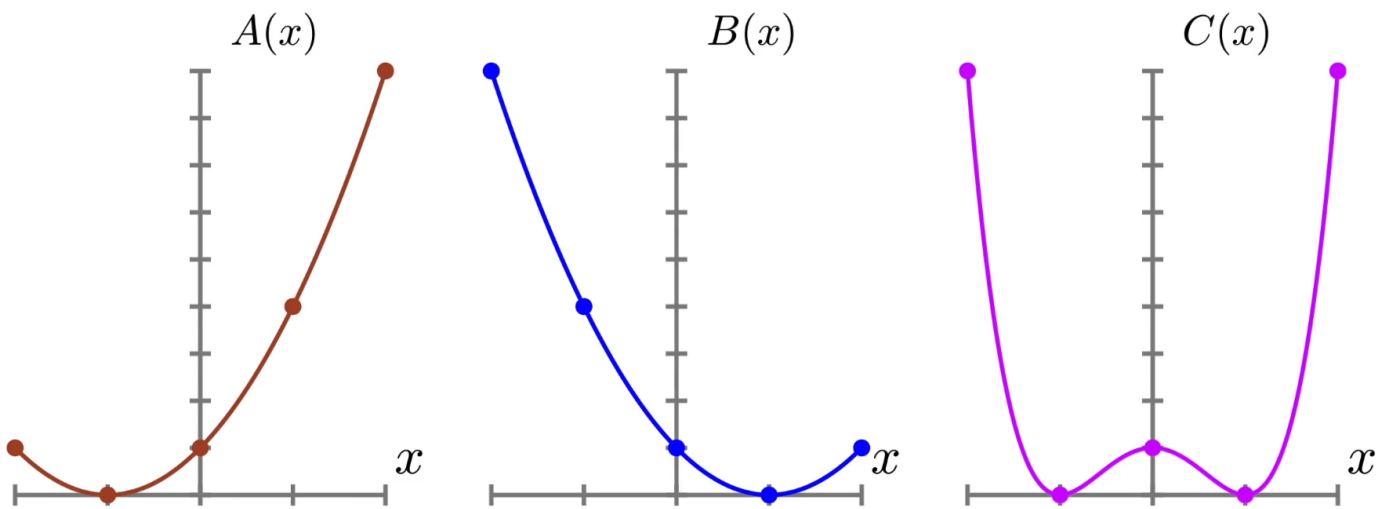
How can this be efficient?!

- Sampling: evaluate the polynomial at $2n+1$ points, $\mathcal{O}(n^2)$
- Pointwise multiplication: $\mathcal{O}(n)$
- Interpolation: up to $\mathcal{O}(n^3)$

$A(x) = x^2 + 2x + 1$ $[(-2,1),(-1,0),(0,1),(1,4),(2,9)]$

$B(x) = x^2 - 2x + 1$ $[(-2,9),(-1,4),(0,1),(1,0),(2,1)]$

$C(x) = x^4 - 2x^2 + 1$ $[(-2,9),(-1,0),(0,1),(1,0),(2,9)]$



The Discrete Fourier Transform (in $\mathbb{C}$)

Sampling

Evaluate the polynomial at the N-th roots of unity $\rightarrow$ take the DFT

$$\mathcal{F}(\omega_k) = \sum_{n=0}^{N} x_n e^{-\frac{2\pi i k}{N} n}$$

Interpolation

Interpolate the points $\rightarrow$ take the inverse DFT

$$\mathcal{F}^{-1}(X_k) = \frac{1}{N} \sum_{n=0}^{N} X_n e^{\frac{2\pi i k}{N} n}$$

Algorithms that calculate the DFT and its inverse are called Fast Fourier Transforms and can achieve $\mathcal{O}(n \log n)$ complexity.

FFT in finite fields

- $a(x)$ and $b(x)$ are polynomials in $\mathbb{F}_2[x]$
- Each coefficient $a_i, b_i \in \mathbb{F}_2$ is a bit (0, 1)

Problem: Finite fields (of characteristic 2) do not admit roots of unity
Solution: use powers of a primitive element
Caveat(s):

- The degree must be a power of 2: $n = 2^\ell$
- Polynomials require a different representation

Case study: the Lin-Chung-Han FFT [LCH14]

Cantor basis

Vector spaces and bases

$\mathbb{F}_2^m$: vector space of m-bit strings over $\mathbb{F}_2$, with m power of 2
Elements of $\mathbb{F}_2^m$ are *vectors*, linear combinations of the *basis vectors*
Basis vectors: $\mathbf{v} = (v_1, \cdots, v_m)$

Canonical basis

$$\mathbf{v} = (1, \alpha, \alpha^2, \cdots, \alpha^{m-1})$$

Cantor basis

$$\mathbf{v} = (v_1, \cdots, v_m)$$

$$v_1 = 1 \qquad v_i^2 + v_i = v_{i-1}$$

Subspaces and vanishing polynomials

$$v_1 = 1 \qquad v_i^2 + v_i = v_{i-1}$$

Subspaces induced by Cantor basis

$$\mathbf{W}_0 := \{0\}, \qquad \mathbf{W}_i := span\{v_0, \cdots, v_{i-1}\}$$

with $\mathbf{W}_0 \subset \mathbf{W}_1 \subset \cdots \subset \mathbf{W}_m = \mathbb{F}_2^m$ and $\dim(\mathbf{W}_i) = i$.

Vanishing polynomials

$$s_i(x) = \prod_{a \in \mathbf{W}_i} (x - a)$$

with $\deg(s_i(x)) = |\mathbf{W}_i| = 2^i$.

Novelpoly basis

Properties of the vanishing polynomials

- linearity: $s_i(x) + s_i(y) = s_i(x + y)$
- recursivity: $s_{i+j}(x) = s_i(s_j(x))$, $s_i(v_{i+j}) = v_j$
- $s_i(v_i) = 1$ for v_i element of the Cantor basis ([CCK+17])
- all v_i are roots of $s_{i+1}(x)$

Novelpoly basis

Cantor basis $\rightarrow$ Subspaces $\rightarrow$ Vanishing polynomials $\rightarrow$ Novelpoly basis

$$X_k(x) = \prod (s_i(x))^{b_i} \qquad \text{with } k = \sum_{i=0}^{\log_2 n} b_i 2^i$$

with $\deg(X_k(x)) = k$.

Polynomial evaluation (sampling)

We start from:

- $f(x) = g(X)$, a polynomial of degree 2^m in novelpoly basis
- $\mathbf{W}_m = \mathbb{F}_{2^m}$, the set of points to be evaluated

We want to get:

- $\{f(x) | x \in \mathbf{W}_m\}$, the evaluation of $f(x)$ at the points in $\mathbf{W}_m$

The LCH FFT

Divide et impera

Recursively split:

- The polynomial: $f_0(X) + f_1(X)X_{2^k}$
- The set of points: $\mathbf{W}_{k-1}, (\mathbf{W}_k \setminus \mathbf{W}_{k-1}) = \{w + v_{k-1} | w \in \mathbf{W}_{k-1}\}$

Get 2 for the price of 1

Evaluate f at $\mathbf{W}_{k-1}$; recall that:

- Elements in novelpoly basis are combinations of vanishing polynomials
- Vanishing polynomials are linear: $s_i(x) + s_i(y) = s_i(x + y)$

Thus the evaluations at $(\mathbf{W}_k \setminus \mathbf{W}_{k-1})$ can be obtained starting from the evaluations at $\mathbf{W}_{k-1}$ with a *butterfly* unit

LCH FFT - example

Consider a polynomial f in novelpoly basis of degree greater or equal than $\deg(s_k(x)) = 2^k$ and split it in half:

$$f(x) = p_0(x) + s_k(x)p_1(x)$$

Evaluate $p_0(x)$ and $p_1(x)$ at $\mathbf{W}_{k-1} = \{\alpha \mid \alpha \in \mathbf{W}_{k-1}\}$:

$$p_0(\alpha) + s_k(\alpha)p_1(\alpha)$$

Evaluate $p_0(x)$ and $p_1(x)$ at $(\mathbf{W}_k \setminus \mathbf{W}_{k-1}) = \{\alpha + v_k \mid \alpha \in \mathbf{W}_{k-1}\}$

$$p_0(\alpha + v_k) + s_k(\alpha + v_k)p_1(\alpha + v_k)$$

$p_0(x)$ and $p_1(x)$ are combinations of vanishing polynomials $s_i(x)$ with $i < k$:

$$\sum s_i(\alpha + v_k) = \sum s_i(\alpha) + s_i(v_k) = \sum s_i(\alpha) + 0 = \sum s_i(\alpha)$$

LCH FFT - example

From this consideration, we can observe that:

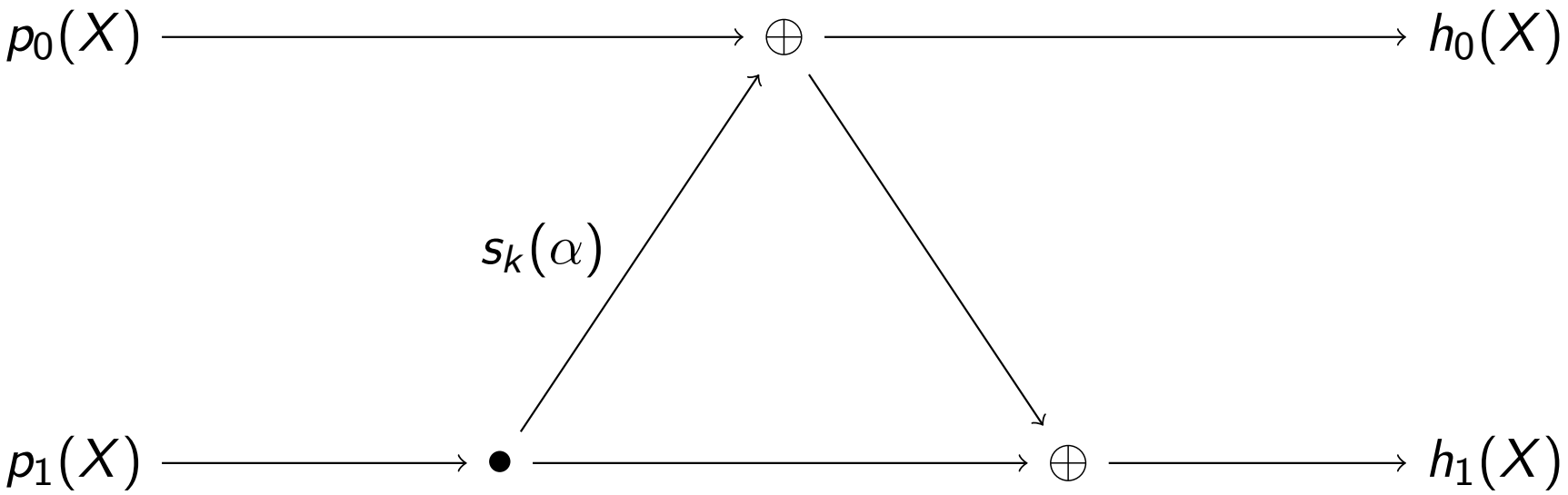
$$p_0(\alpha + v_k) = p_0(\alpha), \qquad p_1(\alpha + v_k) = p_1(\alpha)$$

Hence we can rewrite the evaluation at $(\mathbf{W}_k \setminus \mathbf{W}_{k-1})$ as:

$$p_0(\alpha + v_k) + s_k(\alpha + v_k)p_1(\alpha + v_k) = p_0(\alpha) + s_k(\alpha)p_1(\alpha) + s_k(v_k)p_1(\alpha)$$

and thus reuse the first evaluation to get the second one.

Figure: Butterfly unit



Frobenius map

We can use the Frobenius map

$$\phi : \mathbb{F}_2 \to \mathbb{F}_2 \qquad x \mapsto x^2$$

to derive more values of a polynomial f of degree n from less evaluations.

Frobenius partition

We can build a set of points $\mathbf{S} \subset \mathbb{F}_{2^m}$ as:

$$\mathbf{S} := v_{I+m/2} + \mathbf{W}_I = \{v_{I+m/2} + x \mid x \in \mathbf{W}_I\}$$

which has size $n' = |\mathbf{S}| = |\mathbf{W}_I| = n/m$. By repeatedly applying ϕ to $\mathbf{S}$ we get

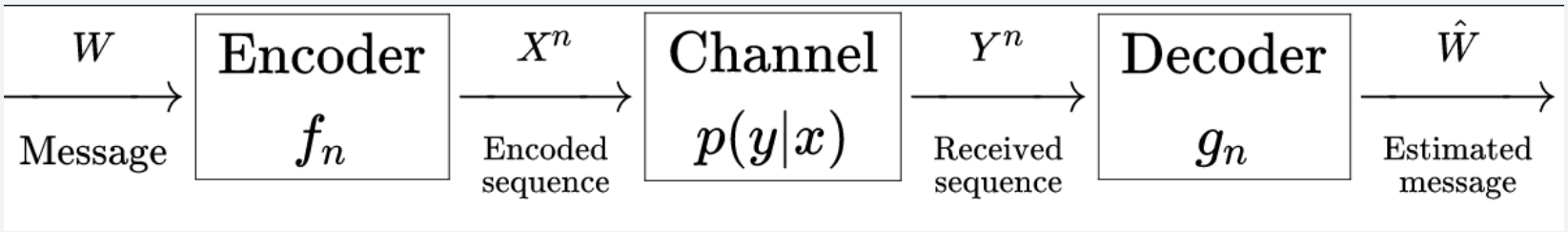
$$\mathbf{S} \cup \phi(\mathbf{S}) \cup \cdots \cup \phi^{m-1}(\mathbf{S}) = \mathbf{O}$$

with $|\mathbf{O}| = n$.

Error Correction Codes

Problem: we want to send a message m over a noisy channel.

- The channel is noisy: it introduces errors in the message
- The receiver wants to recover the original message



Solution: send, along with the message, information that allows the receiver to correct the errors.

Linear codes

Let $\mathcal{V}$ be a vector space of dimension n over $\mathbb{F}_2$. A linear code $\mathcal{C} = [n, k]$ is a subspace of $\mathcal{V}$ of dimension k.

- Messages are k bits long
- We add $n - k$ bits to detect and correct errors (up to δ)

Generator Matrix

The matrix $\mathbf{G} \in \mathbb{F}_2^{k \times n}$ is the generator matrix of $\mathcal{C}$ if

$$\mathcal{C} = \{\mathbf{mG} \mid \mathbf{m} \in \mathbb{F}_2^k\}$$

Syndrome

Parity Check Matrix

The matrix $\mathbf{H} \in \mathbb{F}_2^{(n-k)\times n}$ is the parity check matrix of $\mathcal{C}$ if

$$\mathcal{C} = \{\mathbf{v} \in \mathbb{F}_2^n \mid \mathbf{H}\mathbf{v}^\top = 0\}$$

The syndrome of a word $\mathbf{v} \in \mathcal{C}$ is defined as:

$$\mathbf{S}(\mathbf{v}) = \mathbf{H}\mathbf{v}^\top$$

This definition is equivalent to the one based on the generator matrix.

Syndrome Decoding

Given an encoded message, find the closest message w.r.t. a certain metric (*i.e.* Hamming weight, $\mathtt{wt}()$).

How hard can this be?

- Take a word $\mathbf{x}$ with $\mathtt{wt}(w)$
- Calculate the syndrome $\mathbf{y}^\top = \mathbf{H}\mathbf{x}^\top$
- Given $\mathbf{H}$ and $\mathbf{y}^\top$, retrieve $\mathbf{x}$

This problem is **NP-complete** [BMvT78].

Hamming Quasi-Cyclic

- IND-CPA Public Key Encryption
- IND-CCA2 Key Encapsulation Mechanism
- Based on a variant of the syndrome decoding problem
- Employs quasi-cyclic codes to shorten key sizes

Codes

HQC employs two codes:

- a decodable $\mathcal{C} = [n, k]$ generated by $\mathbf{G} \in \mathbb{F}_2^{k \times n}$ (public)
- a random $[2n, n]$ code with parity-check matrix $\mathbf{H} = (\mathbf{I}_n | \mathbf{circ}(\mathbf{h}))$

Public Key Encryption

Setup

Generate the parameters $(n, k, \delta, w, w_r, w_e)$ for the corresponding security level λ

Key Generation

- Generate $seed_h \xleftarrow{\$} \{0,1\}^\lambda$
- Generate $\mathbf{h} \xleftarrow{\mathrm{PRNG}(seed_h)} \mathbb{F}_2^n$
- Generate $(\mathbf{x}, \mathbf{y}) \xleftarrow{\$} \mathbb{F}_2^n$, with $\mathtt{wt}(\mathbf{x}) = \mathtt{wt}(\mathbf{y}) = w$
- Calculate $\mathbf{s} = \mathbf{x} + \mathbf{hy}$

$$\mathsf{pk} = (\mathbf{h}, \mathbf{s}) \qquad \mathsf{sk} = (\mathbf{x}, \mathbf{y})$$

Public Key Encryption

To encrypt a message $\mathbf{m} \in \mathbb{F}_2^k$:

Encryption

- Generate $(\mathbf{r_1}, \mathbf{r_2}) \xleftarrow{\$} \mathbb{F}_2^n$, with $\mathtt{wt}(\mathbf{r_1}) = \mathtt{wt}(\mathbf{r_2}) = w_r$
- Generate $\mathbf{e} \xleftarrow{\$} \mathbb{F}_2^n$, with $\mathtt{wt}(\mathbf{e}) = w_e$
- Calculate $\mathbf{u} = \mathbf{r_1} + \mathbf{h}\mathbf{r_2}$
- Calculate $\mathbf{v} = \mathbf{m}\mathbf{G} + \mathbf{s}\mathbf{r_2} + \mathbf{e}$

$$\mathrm{ctx} = (\mathbf{u}, \mathbf{v})$$

Public Key Encryption

Decryption

From the ctx $= (\mathbf{u}, \mathbf{v})$ and the private key pk $= (\mathbf{x}, \mathbf{y})$:

$$\begin{aligned}\mathbf{v} - \mathbf{u} \cdot \mathbf{y} &= (\mathbf{mG} + \mathbf{s} \cdot \mathbf{r_2} + \mathbf{e}) - (\mathbf{r_1} + \mathbf{h} \cdot \mathbf{r_2})\mathbf{y} = \mathbf{mG} + \mathbf{x} \cdot \mathbf{r_2} - \mathbf{r_1} \cdot \mathbf{y} + \mathbf{e} \\ &= \mathbf{mG} + (\mathbf{x} + \mathbf{h} \cdot \mathbf{y})\mathbf{r_2} + \mathbf{e} - (\mathbf{r_1} \cdot \mathbf{y} + \mathbf{h} \cdot \mathbf{r_2} \cdot \mathbf{y}) \\ &= \mathbf{mG} + \mathbf{x} \cdot \mathbf{r_2} + \mathbf{h} \cdot \mathbf{y} \cdot \mathbf{r_2} + \mathbf{e} - \mathbf{r_1} \cdot \mathbf{y} - \mathbf{h} \cdot \mathbf{r_2} \cdot \mathbf{y} \\ &= \mathbf{mG} + \mathbf{x_2} - \mathbf{r_1} \cdot \mathbf{y} + \mathbf{e}\end{aligned}$$

So we have $\mathbf{m} = \mathcal{C}.\text{Decode}(\mathbf{v} - \mathbf{u} \cdot \mathbf{y})$ whenever

$$\mathtt{wt}(\mathbf{x} \cdot \mathbf{r_2} - \mathbf{r_1} \cdot \mathbf{y} + \mathbf{e}) \leq \delta$$

Key Encapsulation Mechanism

By applying the Fujisaki-Okamoto transformation it is possible to build a IND-CCA2 KEM. Let $\mathcal{G}, \mathcal{H}, \mathcal{K}$ be hash functions and $\mathcal{E}$ an instance of HQC.PKE

- **Setup**: generate the parameters λ as in HQC.PKE, except that k will be the length of the key to be exchanged
- **KeyGen**: as in HQC.PKE

Key Encapsulation Mechanism

Let $\mathcal{G}, \mathcal{H}, \mathcal{K}$ be hash functions and $\mathcal{E}$ an instance of HQC.PKE

Encapsulation

- Generate the seed $\mathbf{m} \xleftarrow{\$} \mathbb{F}_2^k$ and the randomness $\theta \leftarrow \mathcal{G}(\mathbf{m})$
- Generate the ciphertext $\mathbf{c} \leftarrow (\mathbf{u}, \mathbf{v}) = \mathcal{E}.\text{Encrypt}(\text{pk}, \mathbf{m}, \theta)$
- Derive the symmetric key $K = \mathcal{K}(\mathbf{m}, \mathbf{c})$
- Calculate $\mathbf{d} \leftarrow \mathcal{H}(\mathbf{m})$
- Send the pair $(\mathbf{c}, \mathbf{d})$

Key Encapsulation Mechanism

Let $\mathcal{G}, \mathcal{H}, \mathcal{K}$ be hash functions and $\mathcal{E}$ an instance of HQC.PKE

Decapsulation

- Decrypt $\mathbf{m}' \leftarrow \mathcal{E}.\text{Decrypt}(\text{sk}, \mathbf{c})$
- Compute $\theta' \leftarrow \mathcal{G}(\mathbf{m}')$ and $\mathbf{c}' \leftarrow \mathcal{E}.\text{Encrypt}(\text{pk}, \mathbf{m}', \theta')$
- Check that $\mathbf{c} = \mathbf{c}'$ and $\mathbf{d} = \mathcal{H}(\mathbf{m}')$
- Derive the shared key $K \leftarrow \mathcal{K}(\mathbf{m}, \mathbf{c})$

Implementation

Parameters

Our target: `hqc-128` on ARM Cortex-M4

Cryptosystem parameters

- Maximum polynomial degree: $n = 17669$
- Hamming weight of sparse polynomials: $\omega = 75$

Target board

STM32L476RG

- 32-bit CPU
- 80 MHz
- 128 KB RAM
- 1 MB Flash

Polynomial representation

We consider the ring of binary polynomials $\mathcal{R} = \mathbb{F}_2[x]/\langle x^n + 1 \rangle$

We have two ways of representing binary polynomials:

- sparse: a vector of indexes, every index is a non-null coefficient
- dense: a vector of coefficients, every entry is a coefficient

Example

The polynomial $f(x) = x^{15} + x^{13} + x^8 + x^6 + x + 1$ can be represented as:

- sparse: $[15, 13, 8, 6, 1, 0]$
- dense: $[1, 0, 0, 0, 0, 0, 1, 0, 1, 0, 0, 0, 1, 0, 1, 1]$

Memory usage - 32 bit architectures

- sparse: one integer per coefficient (*i.e.* 4 bytes per coefficient)
- dense: one bit per coefficient (*i.e.* 0.25 bytes per coefficient)

Shift&Add

Multiplying a dense polynomial by x^i is equivalent to a circular shift ($\lll_n$) of i positions to the left:

- $a(x)$ in dense representation
- $b(x) = [b_0, \cdots, b_{\omega-1}]$ in sparse representation
- Calculate $c_i(x) \leftarrow a(x) \lll_n b_i$
- Sum the partial results $c(x) = \sum_{i=0}^{\omega-1} c_i(x)$

Pros and cons

- Extremely fast for $\omega \ll n$
- Non constant-time execution

Constant time S&A

Calculate all the shifts, but only sum the ones that are needed:

Monomial-Polynomial multiplication

Let $a(x)$ be a dense polynomial and x^b a monomial, with $b = (\beta_{m-1}\beta_{m-2}\cdots\beta_0)_{\texttt{bin}}$. For each bit $i \in \{0, \cdots, m-1\}$, we have:

- $tmp(x) \leftarrow a(x) \lll_n i$
- if $\beta_i = 1$ then $\texttt{mask} = [111\cdots11]_n$, otherwise $\texttt{mask} = [000\cdots00]_n$
- $c(x) \leftarrow c(x) + ((tmp(x) \wedge \texttt{mask}))$
  - $\beta_i = 1$: $c(x) \leftarrow c(x) + tmp(x)$
  - $\beta_i = 0$: $c(x) \leftarrow c(x) + [000\cdots00]_n$

Pros and cons

- Constant-time execution
- Slower than plain S&A

Frobenius Additive FFT

Based on the BIKE optimized implementation on ARM Cortex-M4 [CCK21]

Let $a(x)$ and $b(x)$ be two dense polynomials of degree n a power of 2

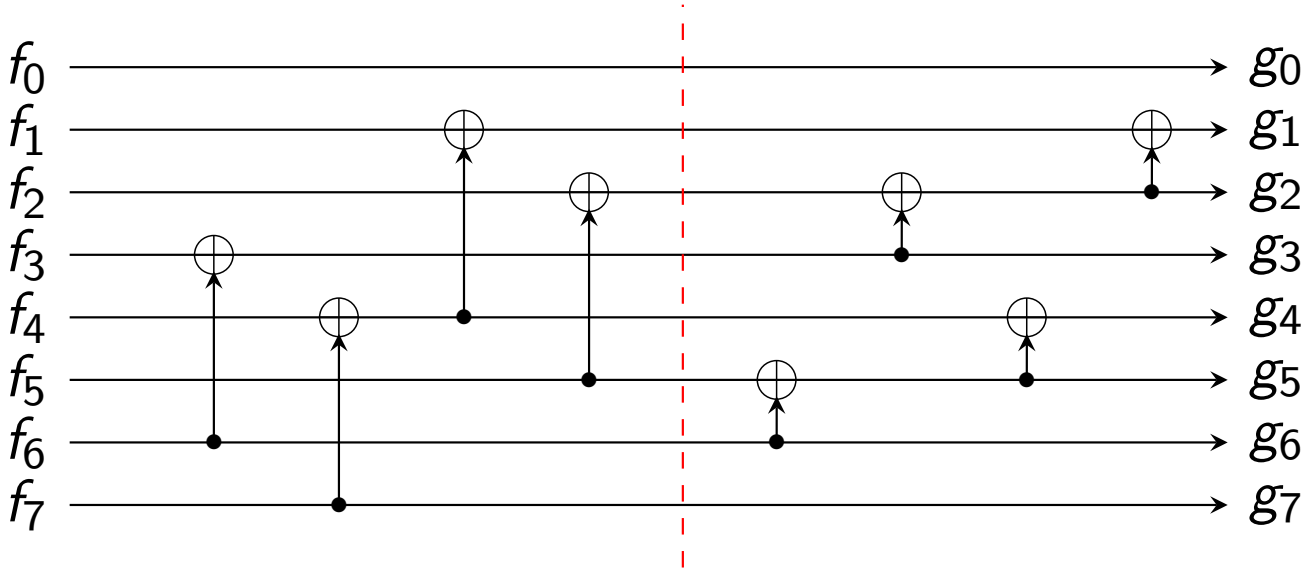
- Basis conversion (from monomial to novelpoly)
- Evaluation with the FFT
  - Encode (truncated FFT)
  - Butterflies
- Pointwise multiplication
- Inverse FFT
- Basis conversion (from novelpoly to monomial)

Inverse operations follow the same principles of the forward ones.

Operands have fixed sizes $\rightarrow$ unroll and optimize recursive calls

Basis conversion

Recursively split the polynomial into halves by dividing by $s_i(x)$
No actual division required, just XORs:



Also, skip the (padded) null coefficients

Evaluation with the FFT

Frobenius partition

Polynomial degree $2 \cdot 17669 \rightarrow n = 65536$

32-bit words $\rightarrow m = 32$

Optimal Frobenius partition:

$$\mathbf{S} = v_{l+m/2} + \mathbf{W}_l = v_{27} + \mathbf{W}_{11}$$

with $l = \log_2(n/m)$

Encoding

The first $\log_2(m)$ layers of the FFT use multipliers depending only on m

- Encode (truncated FFT) the first $\log_2(m)$ recursion layers
- *Classic* butterflies for the remaining layers

Encoding

The first $l_m = \log_2(m)$ layers of the FFT convert elements from $\mathbb{F}_2^n$ to $\mathbb{F}_{2^m}^{n'}$. The twiggle factors can be rewritten as

$$s_{l+1}(v_{l+m/2}), \cdots, s_{l+l_m}(v_{l+m/2})$$
$$v_{m/2-1}, \cdots, v_{m/2-l_m}$$

so the encoding can be implemented as a linear map:

$$g_i = \sum_{j=0}^{m-1} f_{j \cdot n' + i} \cdot r_j = \begin{bmatrix} f_{0 \cdot n'} & f_{1 \cdot n'} & \cdots & f_{(m-1) \cdot n'} \end{bmatrix} \cdot \begin{bmatrix} r_0 \\ r_1 \\ \vdots \\ r_{m-1} \end{bmatrix}$$

Multiplications

Butterfly twiggle factors

Multiplication step: $\beta \cdot p_1(s_1(\beta))$, $\beta = (v + w) \in v_{l+m/2-i} + \mathbf{W}_{l-i}$

- Precompute and store all the β values
- Notice that $\gamma \cdot \beta = \gamma \cdot (v + w) = \gamma \cdot v + \gamma \cdot w$
  - $\gamma \cdot v$ can be computed with a binary circuit (v are fixed)
  - $\gamma \cdot w$ can be computed in a subfield to reduce the number of bits

Pointwise multiplications

The resulting evaluation points are, by construction, in $\mathbb{F}_{2^{32}}$.
Apply Karatsuba (only once) to operate in $\mathbb{F}_{2^{16}}$ and have the results fit in the 32-bit CPU registers.

Experimental results

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